Ateneo de Davao University

Computer Studies Cluster

CS 3145

SIA 1: Software Engineering

First Semester 2025 - 2026

Davao Blue Eagles Marching Band

Proponents:

Harley Love N. Cuba
Jan-Aceryl N. Futalan
Ivan Louie L. Malicsi

Adviser: Jose Mari Freires

Panelist:
Bernie Jereza
Dwight De Jesus

Business Background

Business Name	Davao Blue Eagles Marching Band
Address	Matina Crossing, Davao City
Type of Business	Marching Band
Link (Site, FB, Maps, etc.,)	https://www.facebook.com/DBEMB2012
Background Information	The Davao Blue Eagles Marching Band (DBEMB) is a musical organization based in Matina Crossing, Davao City. Established on November 24, 2012, the band gained recognition after winning its first major competition in Bohol. Over the years, it has competed and earned multiple championship titles across the Philippines. After the pandemic, DBEMB faced challenges such as reduced membership and financial difficulties but continues to rebuild and promote musical excellence in the community.

System Architecture

Main Framework	The system is developed using JavaScript with JSX syntax as the primary programming language. The front-end is built with React.js, a powerful JavaScript library for building interactive user interfaces through reusable components. For navigation and routing, the application utilizes React Router DOM to handle customer-side routing seamlessly. Unlike many modern web applications that rely on CSS frameworks like Bootstrap or Tailwind CSS, this system employs a custom styling approach using CSS-in-JS methodology, where styles are defined as JavaScript objects directly within components, providing better component isolation and dynamic styling capabilities. The back-end infrastructure is powered by Node.js with Express.js, a minimal and flexible web application framework that handles RESTful API endpoints. The server implements CORS (Cross-Origin Resource Sharing) middleware to enable secure cross-origin requests between the front-end and back-end. For data persistence, the system uses MySQL, a robust relational database management system. Authentication and security are handled through bcryptjs, which provides industry-standard password hashing and validation to protect user credentials.
GitHub Repo	https://github.com/illmalicsi/SAD-Final
Live Deployment (if exist)	
Plugins/Packages Used	The development of this system leverages various third-party packages and plugins to enhance functionality and streamline development. On the front-end, react-icons is utilized to provide a comprehensive icon library including Font Awesome and Bootstrap icons throughout the user interface. React Router DOM manages all customer-side navigation and routing, enabling a smooth single-page application experience. The

	application also implements custom service modules including authService for authentication operations, notificationService for managing user notifications, and billingService for handling payment-related functionality. The back-end relies on several critical packages to ensure robust server operations. Express.js serves as the core web framework, handling HTTP requests and routing. The mysql2 package provides a modern MySQL database driver with Promise support, enabling efficient asynchronous database operations. Security is paramount, so bcryptjs is employed for password encryption and validation, ensuring that user credentials are never stored in plain text. The cors package enables controlled cross-origin resource sharing, while .env manages environment variables securely. Body-parser middleware processes incoming request bodies, and multer handles file uploads for features like identity proof verification and user avatar uploads. For development efficiency, the project uses nodemon to automatically restart the server when code changes are detected, significantly improving the development workflow. Package management is handled through npm or yarn, ensuring consistent dependency management across different development environments.
Database System	The system utilizes MySQL as its database platform, a widely-adopted open-source relational database management system (RDBMS) that provides powerful, reliable, and scalable data storage capabilities. MySQL serves as the backbone for all data persistence, offering robust features including ACID (Atomicity, Consistency, Isolation, Durability) compliance, efficient query optimization, and excellent performance for both read and write operations. The database architecture follows a well-structured relational model with multiple interconnected tables that maintain referential integrity through foreign key relationships. Core tables include users for managing account information across members, customers, and administrators; bookings for tracking service reservations and instrument rentals; instrument_inventory for cataloging available instruments; instrument_locations for real-time instrument tracking and availability management; notifications for the user notification system; payments and transactions for financial record-keeping; and membership_applications for processing new member registrations. To ensure optimal performance, the database implements strategic indexing on frequently queried columns such as booking_date, status, and user_id, significantly improving query response times. Connection efficiency is maintained through mysql2's connection pooling feature, which manages database connections intelligently, reducing overhead and improving overall system performance. This robust database architecture ensures data integrity, scalability, and reliability as the system grows and handles increasing user loads.

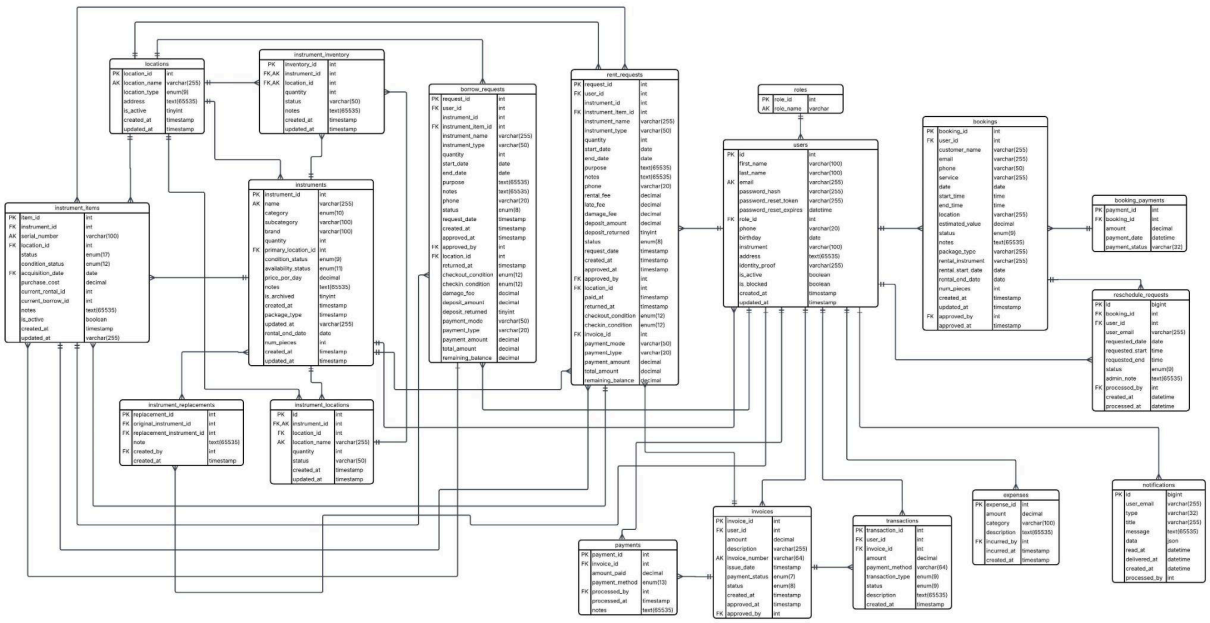
Approved Modules

Module	Sub-Modules / Functions	Description / User Story
(1) User Management	Login / Logout	As a user, I want to log in and log out of the website, so that I can securely access my personal account and protect my information when I'm done using it.
	Customer Signup	As a new customer, I want to create an account on the website so that I can access personalized features and manage my profile securely.
	Block/Unblock account	As an admin, I can block or unblock inactive users to maintain system integrity and ensure active user engagement.
	Membership Signup	As a user, I want to sign up to become a band member by filling out a registration form on the website
	Application Approval	As an admin, I want to approve or reject membership applications.
	Forgot Password	As a user, I want to receive a password reset link through email so I can regain access to my account if I forget my password.
	Reset Password	As a user, I want to update my password through the link sent to my email so I can restore access to my account with a new password.
	Account Creation	As an admin, I want to create new accounts and adjust their roles—whether user, member, or admin—so I can manage system access effectively.
(2) Services and Booking Management	Services Booking Form	As a customer, I want to submit a service booking request.
	View Booking Requests	As an admin, I want to view all service booking requests with complete details.
	Approve/Reject Booking	As an admin, I want to approve or reject service bookings and notify the customer upon login or when the booking status is updated.
	Check Schedule Conflicts	As an admin, I want to be alerted if a new booking request conflicts with an already approved event.

	Cancel Booking	As a customer, I want to cancel my booking.
	Reschedule Booking	As a customer, I want to reschedule my booking if there's an emergency.
	Upcoming and Future Schedules	As an admin, I want to see upcoming and pending bookings for us to prepare.
(3) Instrument Inventory Management	Add Instruments	As an admin, I want to add instruments to the inventory.
	Instrument Borrow	As a member, I want to request to borrow an instrument from the inventory so that I can use it for practice or performances.
	Instrument Rental	As a customer, I want to request to rent an instrument from the inventory.
	Approve/Reject Request	As an admin, I want to review the pending requests and approve or reject instrument requests.
	Return	As an admin, I want to return the rented/borrowed instrument to its specific location.
	Instrument Inventory	As an admin, I want to manage the instrument inventory (add, archive, edit).
(4) Financial Bill Management	Invoice Generation	As an admin, I want to generate invoices for services automatically.
	Payment Processing	As an admin, I want to process payments and update transaction records.
	Transaction History	As a user, I want to view my transaction history and payment status.
(5) General	Notification (user)	As a user, I want to see updates regarding my pending requests (booking, instrument, transaction).
	Notification (admin)	As an admin, I want to see updates regarding pending requests (membership, booking, instrument, transaction).

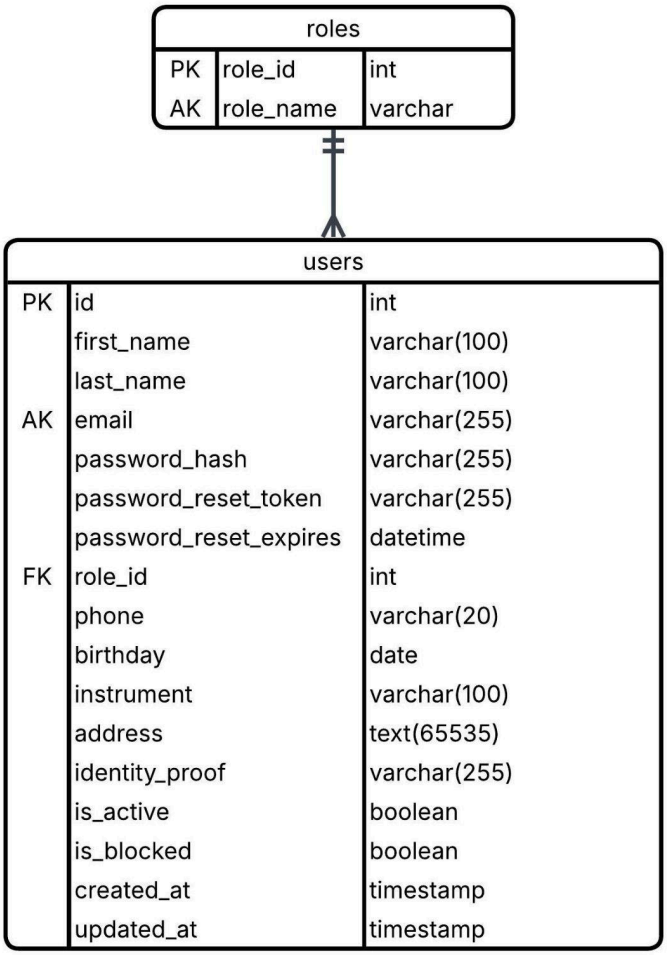
Entity Relationship Diagram

Full Entity Relation Diagram

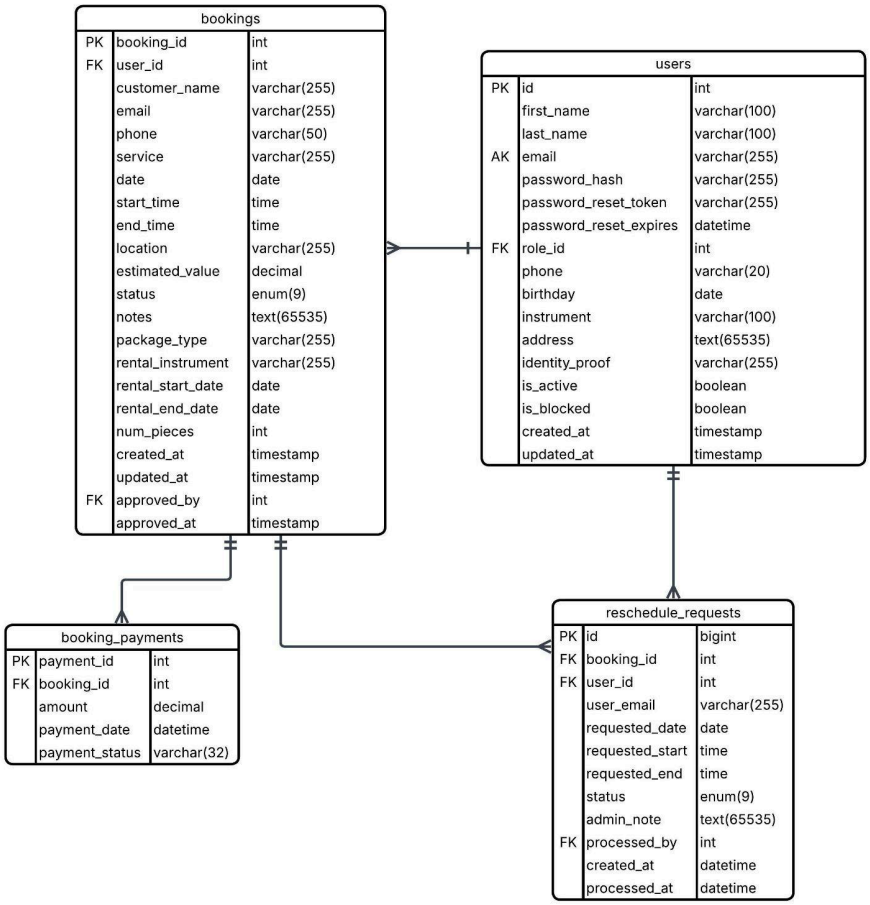


[View Full ERD]

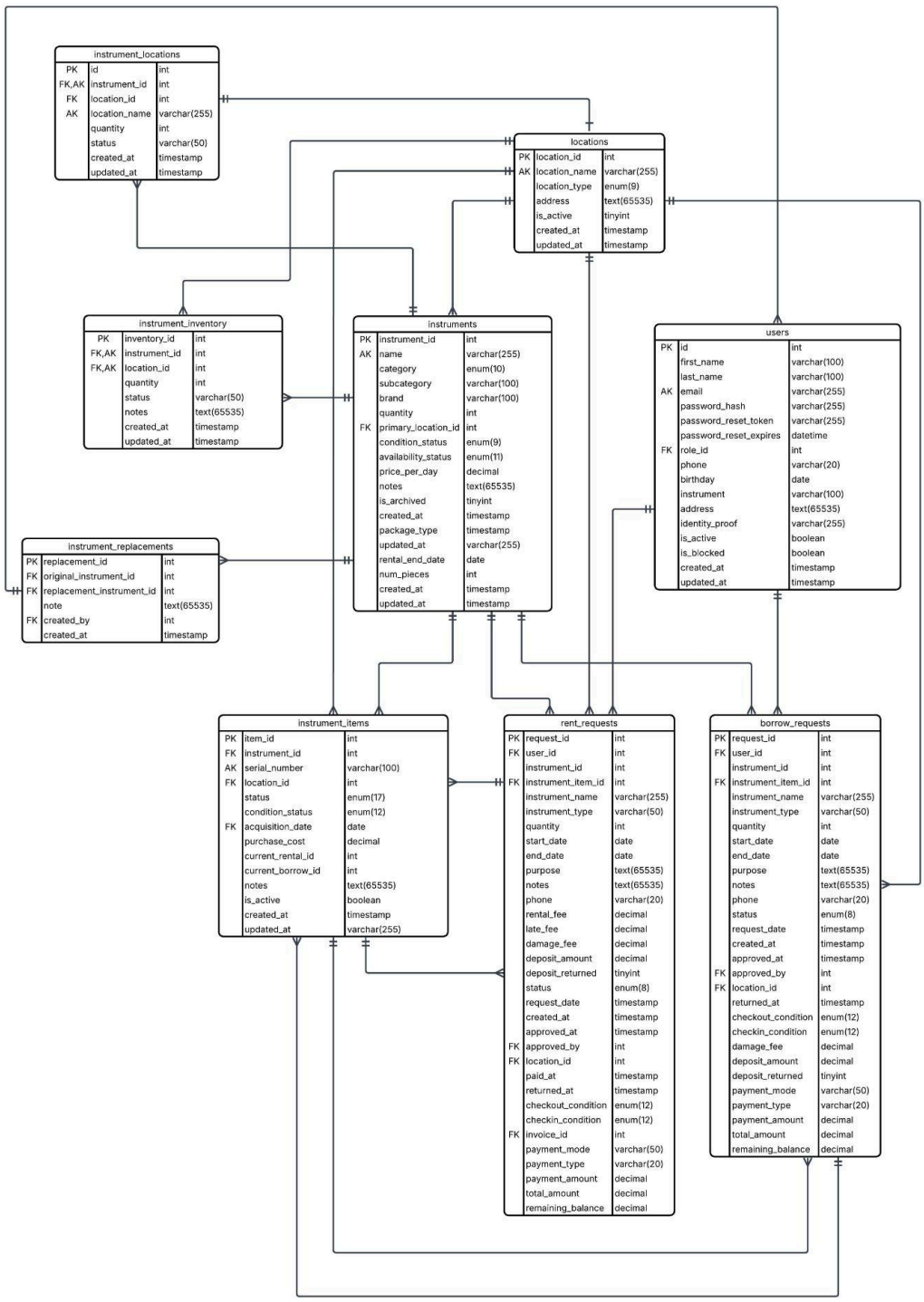
ERD for Module 1: User Management



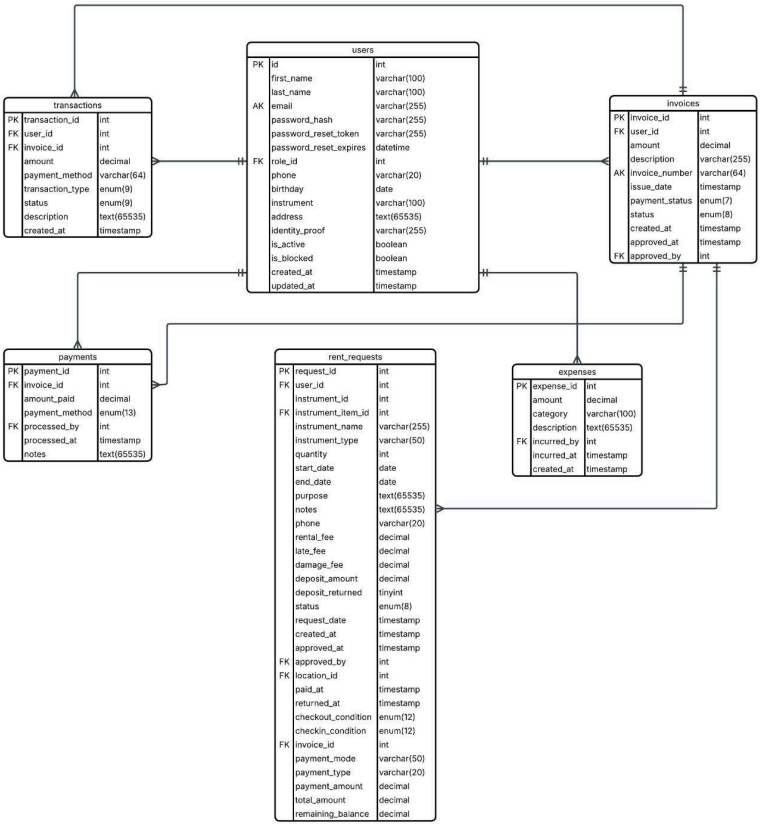
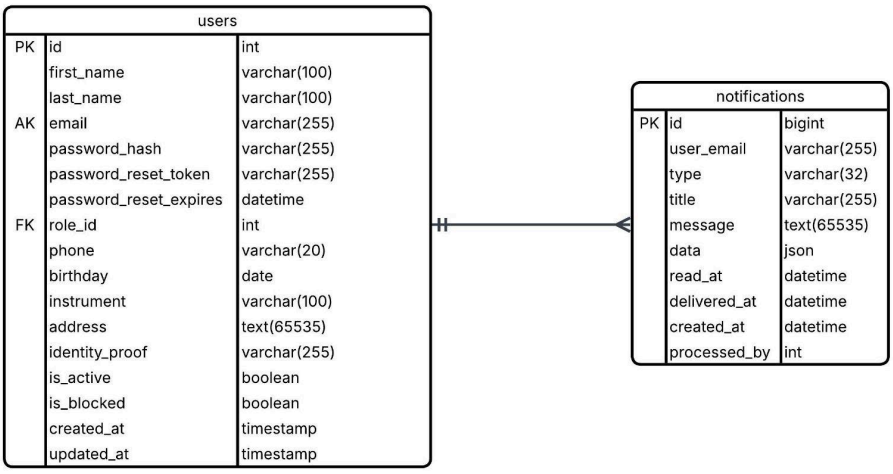
ERD for Module 2: Services and Booking Management



ERD for Module 3: Instrument Inventory Management



ERD for Module 4: Financial Bill Management



ERD for Module 5: General